

# Meeting Notes: Project Progress and Next Steps

Prepared for progress meeting | 10 July 2026

## 1. Where the project is now

**The engineering baseline is now in place.** The main change since the original plan is methodological: ArchEHR-QA remains a useful grounded clinical-QA diagnostic, but it should not be the final evaluation or Semantic Entropy Probe (SEP) training dataset because its released test split does not provide usable gold evidence or answer-quality labels.

- Implemented and tested: multi-sample answer generation, NLI-based semantic clustering, answer Semantic Entropy, token-level uncertainty, citation-set uncertainty, and lightweight evaluation outputs.
- Current focus: choose a replacement dataset with reliable answer-quality/correctness labels, then reuse the existing pipeline.
- SEP / hidden-state probe work is deliberately deferred until the uncertainty target and dataset are stable.

## 2. Work completed since the last update

### 2.1 Semantic Entropy baseline

- A common-schema pipeline was completed for generation, cleaning, semantic clustering, entropy calculation, token scores, and health checks.
- A 50-question PubMedQA server pilot was completed with Qwen2.5-7B-Instruct, five generations per question, CUDA inference, and bidirectional NLI clustering.
- The maintained evaluator now produces majority-vote accuracy, AUROC for error detection, and rejection curves, alongside raw generation and score artifacts.

### 2.2 Grounded ArchEHR-QA pipeline

- The ArchEHR-QA path supports grounded patient questions, clinician rewrites, evidence sentences, structured cited answers, citation parsing, and answer/citation uncertainty as separate signals.
- The Gemma 3 12B test run completed 100 examples x 10 generations with valid JSON parsing for all 1,000 outputs.
- Local tests cover loaders, prompts, parsing, clustering, citation uncertainty, evaluation utilities, and static end-to-end fixtures.

## 3. What the early results show

| Observation                               | Evidence so far                                                                                                   | Interpretation                                                                                                            |
|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| SE captures answer-level variation        | On the 50x5 pilot, 38/50 examples formed 2-4 semantic clusters; token uncertainty correlated only weakly with SE. | Keep Semantic Entropy as a separate baseline rather than treating token scores as a substitute.                           |
| Pilot accuracy is modest                  | PubMedQA majority-vote accuracy was 28/50 (0.560); the best error-detection AUROC was about 0.57.                 | This is a working baseline, not evidence of a strong final method.                                                        |
| Stable answers can still be weak          | In the ArchEHR-QA test diagnostic, 40/48 reference-only low-quality examples had answer SE = 0.                   | SE can miss consistent but incomplete or wrong answers; evaluation needs meaningful quality labels.                       |
| Model choice affects format and citations | On a 20x10 comparison, Gemma 3 12B had 2/200 answers with no valid citation versus 56/200 for Qwen2.5-7B.         | Gemma is the cleaner engineering path for structured grounded outputs, though this is not yet a final quality comparison. |

## 4. Dataset decision: why we need to pivot

**The key blocker is not the implementation.** It is the absence of a trustworthy final evaluation target in the available ArchEHR-QA test data.

- ArchEHR-QA dev has clinician answers and sentence-relevance labels, so it is useful for small diagnostic checks.
- ArchEHR-QA test has clinician answers only; it lacks sentence relevance, citation, and answer-quality labels.
- Reference-overlap scores on test are useful for debugging, but they cannot demonstrate factual grounding or provide a credible final AUROC/ECE benchmark.
- PubMedQA has public yes/no/maybe labels, but its short-form output does not match the intended grounded, long-form clinical-QA setting.

## 5. Proposed next steps

1. Select a replacement dataset with public/approved access and clear correctness or answer-quality labels. Preferred fields: question, context/evidence, reference answer, and quality/correctness labels.
2. Add one dataset adapter and a dataset-specific quality evaluator while keeping the current generation, clustering, token-UQ, and artifact layout unchanged.
3. Run the Semantic Entropy baseline with 10 samples per question, then report accuracy, AUROC, calibration where appropriate, and rejection curves against real labels.
4. Inspect failure cases, especially low-SE wrong answers, before deciding whether SEP probes or token-selection experiments are justified.

## 6. Current server issue

**Status: authentication succeeds, but the onward connection is refused.** The SSH debug output confirms public-key authentication to the Isambard login gateway, so the local key and initial login path are working. The failure occurs only when SSH attempts to forward to the Phase 2 access node.

| Stage              | Observed result                                                                                                 |
|--------------------|-----------------------------------------------------------------------------------------------------------------|
| Login gateway      | Authenticated to ai.login.isambard.ac.uk using publickey.                                                       |
| Phase 2 forwarding | ai-p2.access.isambard.ac.uk:22 returns Connection refused.                                                      |
| Implication        | This looks like an access-node/service-side forwarding problem rather than a public-key authentication failure. |

- Practical next action: check the current Phase 2 access instructions/status and send the exact timestamp plus the three SSH lines to the Isambard support team if the problem persists.

## 7. Questions for discussion

- What replacement dataset best preserves the clinical/grounded motivation while providing reliable quality labels?
- Is a well-evaluated Semantic Entropy baseline plus failure analysis a sufficient next milestone before SEP work?
- Should the final target emphasise correctness, evidence support/citation quality, or a combination of these?
- Should we treat the server forwarding problem as a support issue and pause new GPU runs until access is restored?

## 8. Short spoken summary

**Since the last meeting, I have completed the engineering Semantic Entropy baseline and the grounded ArchEHR-QA pipeline.** The server pilots show that the pipeline produces the expected artifacts and that Semantic Entropy captures answer-level variation that is not the same as token uncertainty. However, the current results also show an important limitation: stable answers can still be low quality. The main project decision is therefore to stop treating ArchEHR-QA test as the final benchmark, because it lacks gold evidence and answer-quality labels. The next step is to choose a replacement dataset with usable supervision, reuse the completed pipeline, and then evaluate Semantic Entropy properly before moving on to SEP probes. Separately, the current SSH issue is not an authentication failure: the login gateway accepts the key, but forwarding to the Phase 2 access node is being refused.

### Useful phrasing if asked about the dataset pivot

The code and the uncertainty calculations are not being discarded. The pivot is about making the evaluation defensible: the same generation and Semantic Entropy infrastructure can be reused, but it needs a dataset where an uncertainty score can be checked against an actual answer-quality target.

### Key facts to have ready

- PubMedQA pilot: 50 questions, 250 generations, five samples per question; majority-vote accuracy 0.560.
- ArchEHR-QA Gemma run: 100 questions, 1,000 generations, all outputs parsed as valid JSON.
- ArchEHR-QA limitation: the test release contains clinician answers but no evidence, citation, or answer-quality labels.

### Useful phrasing for the server update

The SSH key is accepted by the login gateway. The problem begins only when the service forwards the session to the Phase 2 access node, which refuses the connection. I will check the access status and raise it with support with the exact debug lines if it continues.